

# **RBM-101: Alkaline-tolerant RNA aptamers useful to purify acid-sensitive antibodies in neutral conditions**

**March, 2020**

# Agenda

1. RIBOMIC and its core technology
2. Anti-IgG aptamer RBM-101:  
Isolation and characterization
3. Application of RBM-101 for Spin  
and LC Columns

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# Corporate Profile

|                          |                                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Company Name:</b>     | <b>RIBOMIC Inc.</b>                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Address:</b>          | <b>3-16-13 Usui Building, Shirokanedai, Minato-ku,<br/>Tokyo 108-0071, Japan</b>                                                                                                                                                                                                                                                                                                                 |
| <b>Date Established:</b> | <b>August 1, 2003</b>                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Date of IPO:</b>      | <b>September 25, 2014 (JPX Mothers code# 4591)</b>                                                                                                                                                                                                                                                                                                                                               |
| <b>Capital:</b>          | <b>JPY 3,261 Million</b>                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Employees:</b>        | <b>20 (including eight Ph.D.)</b>                                                                                                                                                                                                                                                                                                                                                                |
| <b>Business Summary:</b> | <b>Discovery and development of pharmaceutical compounds using<br/>RNA aptamer</b>                                                                                                                                                                                                                                                                                                               |
| <b>Subsidiary:</b>       | <b>RIBOMIC USA Inc. 2120 University Ave Berkeley CA 94704 USA</b>                                                                                                                                                                                                                                                                                                                                |
| <b>Partners :</b>        |  Otsuka  TAISHO PHARMACEUTICAL CO., LTD.  astellas<br><b>Zenyaku Kogyo, Fujimoto Pharmaceutical Corporation,<br/>SEIKAGAKU CORPORATION</b> |
| <b>Collaborations :</b>  | <b>University of Tokyo, Osaka University, Juntendo University,<br/>Osaka Medical College, Chiba Institute of Technology,<br/>Masaryk University (Czech)</b>                                                                                                                                                                                                                                      |

**\*Corporate Profile (As of September 30, 2018)**

# What's Aptamer?

- 10~50 length single strand oligonucleotide (DNA, RNA, modified nucleic acid)
- Formation of 3D structure (depend on the sequence)
- Bind to the target protein strongly like antibody



*Biochimie 2018 vol. 145*

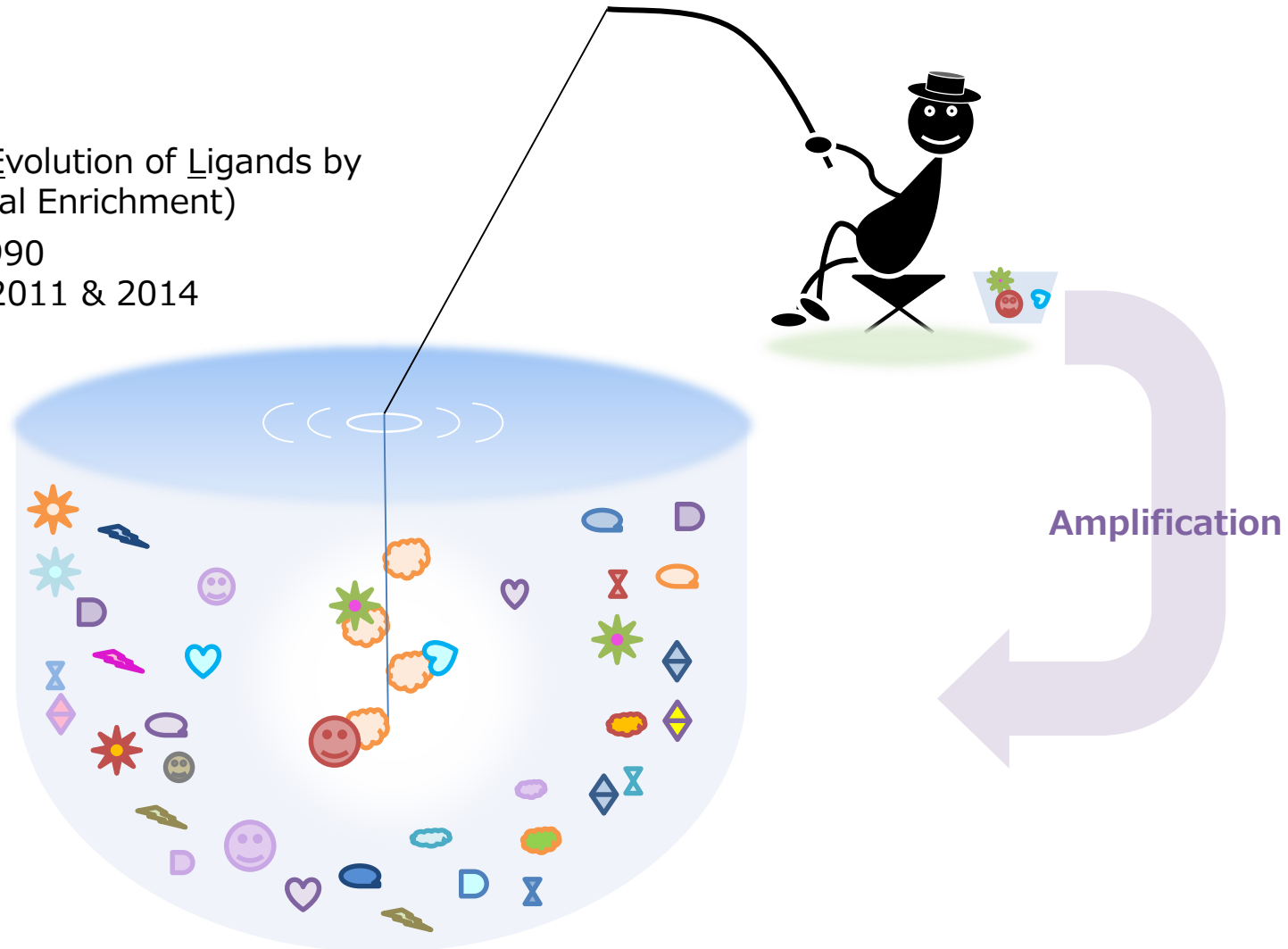
# Selection of Aptamer

## SELEX

(Systematic Evolution of Ligands by Exponential Enrichment)

Larry Gold. 1990

IP expired in 2011 & 2014  
worldwide



Randomized RNA (DNA) libraries ( $\sim 10^{15}$ )

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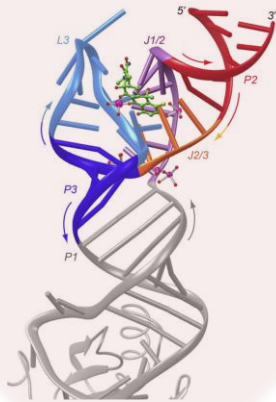
RIBOMIC

# Targets of Aptamer

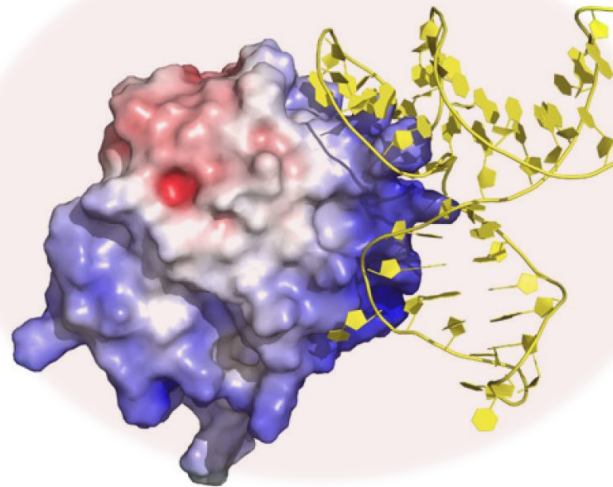
Small  
molecule

Protein

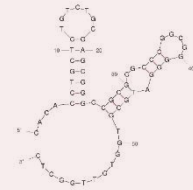
Virus, Cell



*Tetracycline aptamer*  
(ChemBiol 2008 15:1125-37)



*vWF aptamer*  
(Structure 2009 17(11):1476-84)



*E. coli aptamer*

Pathogens 2014 3: 341-355

*Influenza aptamer*



BBRC 2014 443: 37-41

Aptamer can be generated *In vitro* for various targets.

# Product Pipelines

## Therapeutics

| Compound Name    | Target    | Indication                             | Discovery | Pre-clinical | Clinical phase |    |    |   | Partner                                    |
|------------------|-----------|----------------------------------------|-----------|--------------|----------------|----|----|---|--------------------------------------------|
|                  |           |                                        |           |              | I              | 2a | 2b | 3 |                                            |
| RBM-007          | FGF2      | Age-related macular degeneration (AMD) | ■         | ■            | ■              | ■  | ■  |   |                                            |
| RBM-007          | FGF2      | Achondroplasia                         | ■         | ■            |                |    |    |   | AMED (grant support)                       |
| RBM-007          | FGF2      | Pain                                   | ■         | ■            |                |    |    |   |                                            |
| RBM-004          | NGF       | Pain                                   | ■         | ■            |                |    |    |   | Fujimoto Pharm. (licensed)                 |
| RBM-003          | Chymase   | Heart Failure                          | ■         | ■            |                |    |    |   | Osaka Medical College (collaboration)      |
| RBM-010          | ADAMTS5   | Osteoarthritis                         | ■         | ■            |                |    |    |   |                                            |
| RBM-006          | Autotaxin | Fibrosis                               | ■         | ■            |                |    |    |   |                                            |
| RBM-001          | Midkine   | Osteosclerosis                         | ■         | ■            |                |    |    |   |                                            |
| (to be assigned) | IL-21     | Pulmonary Arterial Hypertension        | ■         |              |                |    |    |   | NCCC Osaka (collabo), AMED (grant support) |
| (to be assigned) | ST2       | Severe Asthma, Atopic Dermatitis       | ■         |              |                |    |    |   |                                            |
| (to be assigned) | FGF9      | Psychiatric Disorders                  | ■         |              |                |    |    |   | The Pritzker Consortium                    |

## Affinity Reagents

| Compound Name | Target | Application                                               | Discovery | Product | Commercialization |
|---------------|--------|-----------------------------------------------------------|-----------|---------|-------------------|
| RBM-101       | IgG    | Affinity purification of human IgG and Fc-fusion proteins | ■         | ■       |                   |



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# Profiles of Anti-IgG aptamer, RBM-101

## 1. Affinity purification of human antibodies and Fc-fusion proteins under neutral condition.

- Some antibodies are degenerated and form aggregation during acidic elution step using protein A.

## 2. Durability in alkaline and acidic conditions and culture media.

- Because modified nucleic acids are utilized.

## 3. Specificity to human IgG<sub>1-4</sub>.

- Do not bind to mouse, rat and other IgGs

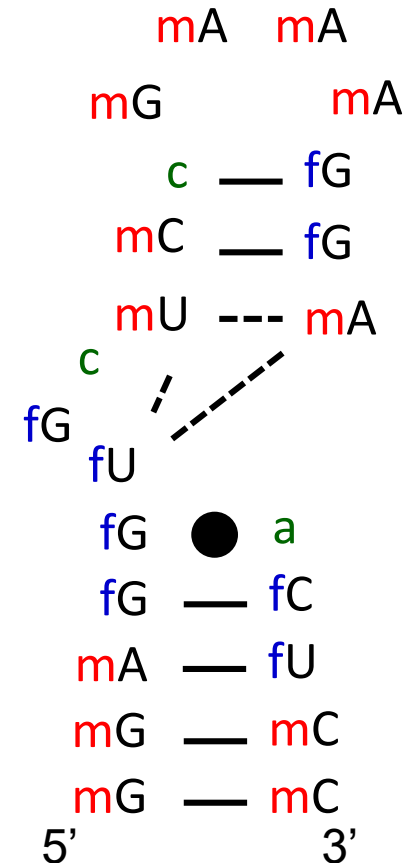
# Characterization of RBM-101

- ▷ Structure
- ▷ Specificity
- ▷ Stability
- ▷ Affinity

# Secondary Structure of RBM-101

- ▶ 23 mer
- ▶ All nucleotides modified at 2' position of ribose
  - m: 2'-OME
  - F : 2'-Fluoro
  - Upper case: RNA
  - Lower case: DNA
- ▶ To increase stability under alkaline condition, all RNA residues have been replaced with modified RNA or DNA in RBM-101.
- ▶ To reduce the synthesis cost, the numbers of raw materials has been minimized.

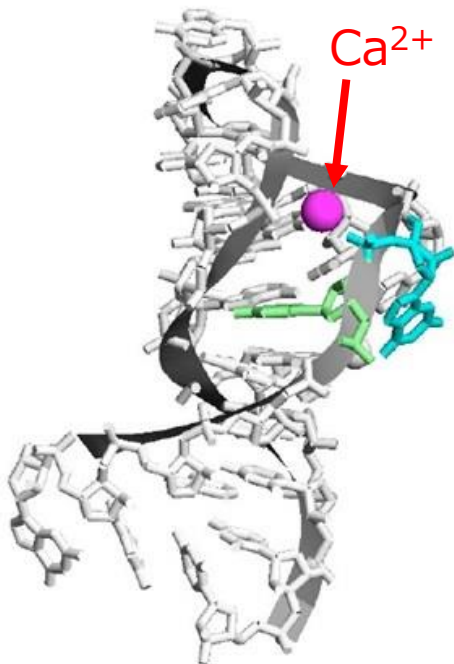
## RBM-101



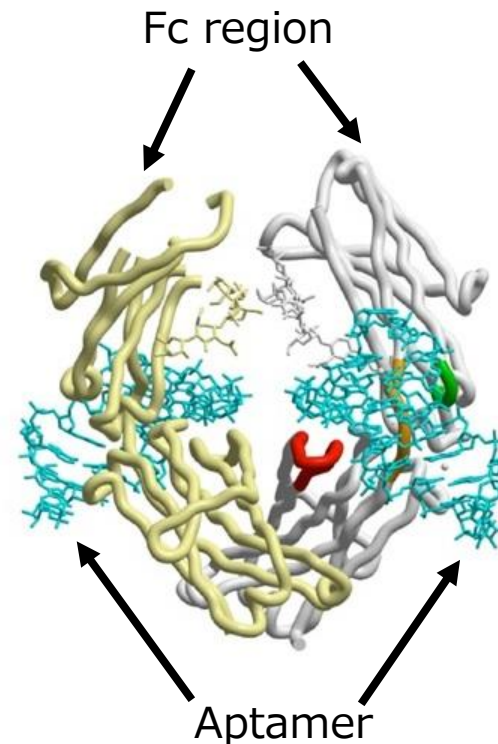
*Biochimie. 2018, 145:113-124*

# Co-Crystal Structure of Aptamer with Fc Region of hIgG<sub>1</sub>

Aptamer



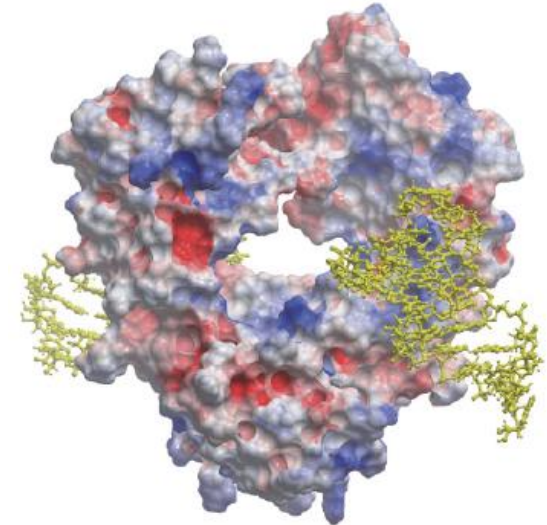
Aptamer – hIgG<sub>1</sub> complex



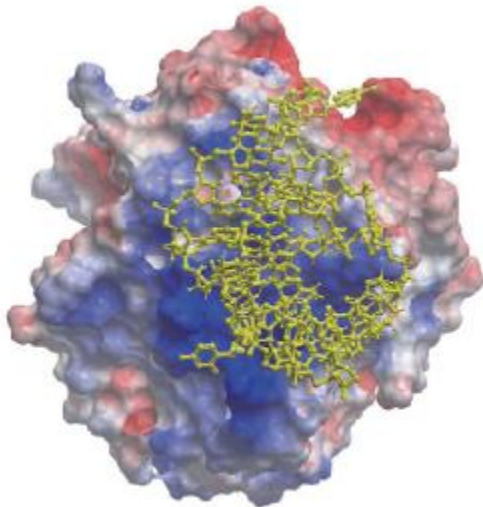
Divalent Cation, such as  $\text{Ca}^{2+}$ , and  $\text{Mg}^{2+}$  is essential for interaction between aptamer and IgG. Therefore, it is easy to inhibit its interaction by removing cations with chelating agents, e.g. EDTA.

# Non-Electrostatic Binding

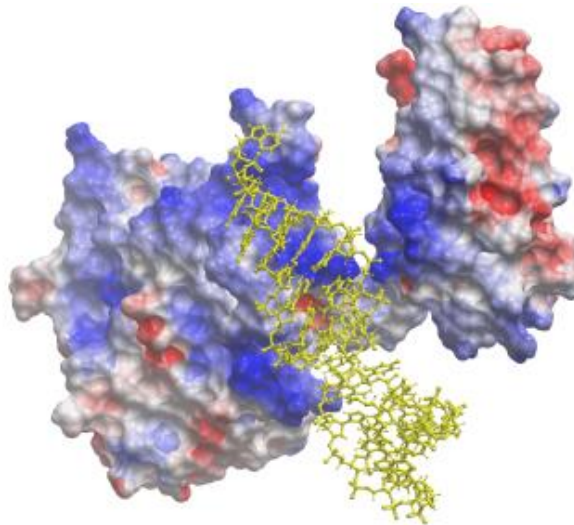
Non-electrostatic interaction as the first case in the aptamer-protein complex.



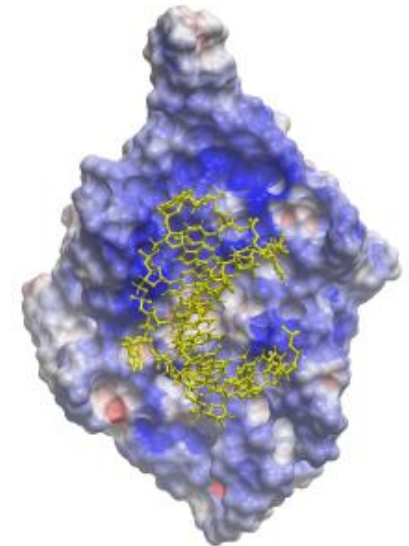
Electrostatic interactions in the other known cases.



aptamer-thrombin complex



Aptamer-NFkb complex

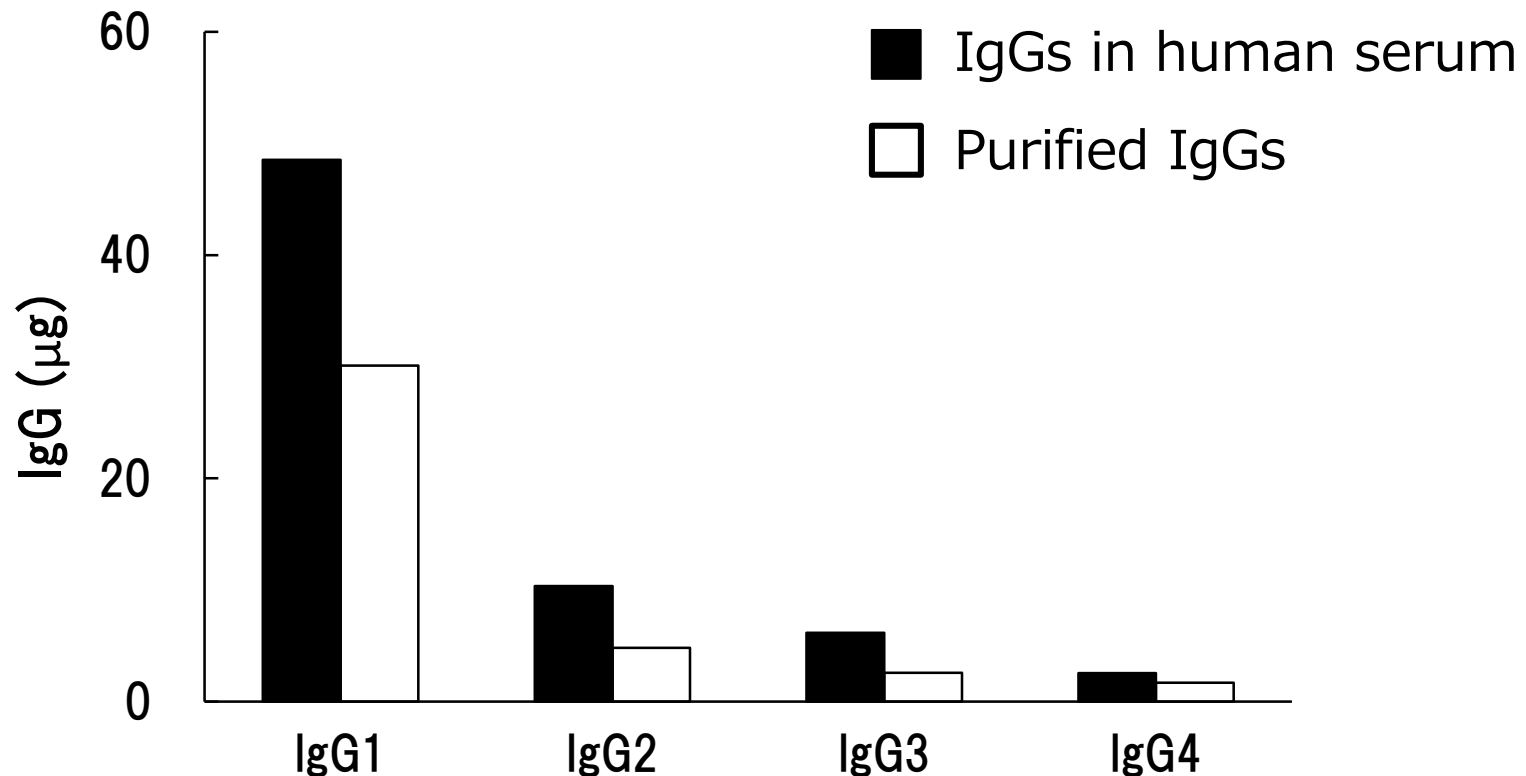


Aptamer-MS2 coat  
protein dimer complex

# RBM101 is Highly Specific to Human IgG<sub>1-4</sub>

|                          | RBM-101 | Protein A                        |
|--------------------------|---------|----------------------------------|
| Human IgG <sub>1-4</sub> | +       | + (except for IgG <sub>3</sub> ) |
| Human IgE                | -       | -                                |
| Human IgA                | -       | -                                |
| Mouse IgG                | -       | +                                |
| Rat IgG                  | -       | +                                |
| Rabbit IgG               | -       | +                                |
| Bovine IgG               | -       | +                                |

# All 4 hIgGs can be purified with IgG aptamer

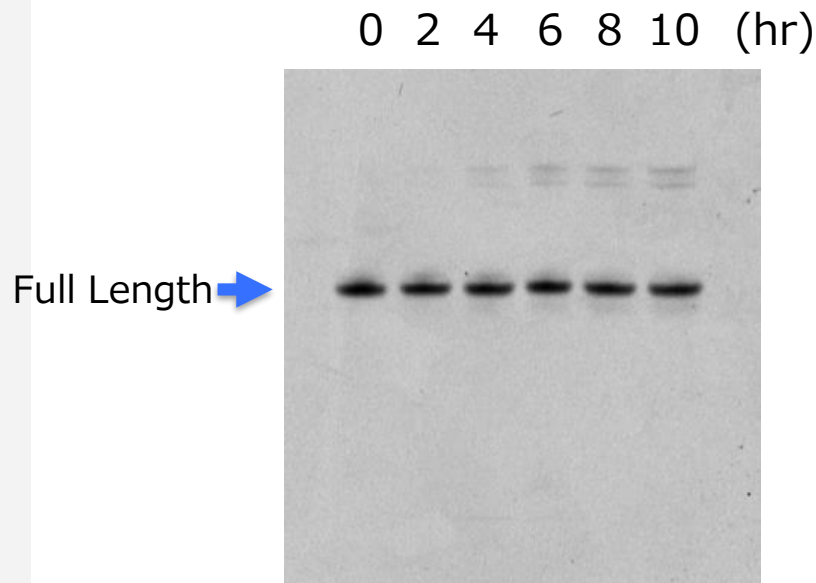


If the sample contains less IgGs than the capacity of aptamer resin, IgG<sub>1</sub>, IgG<sub>2</sub>, IgG<sub>3</sub>, and IgG<sub>4</sub> can be purified equally.



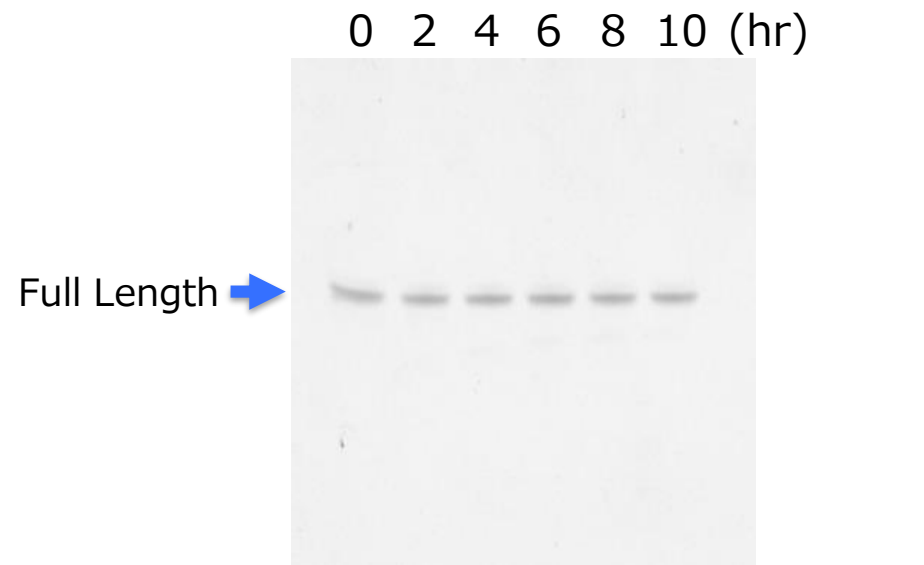
# RBM-101 is Stable under Acidic Condition and in Culture Media

## Under acidic condition



0.1N HCl at 25°C

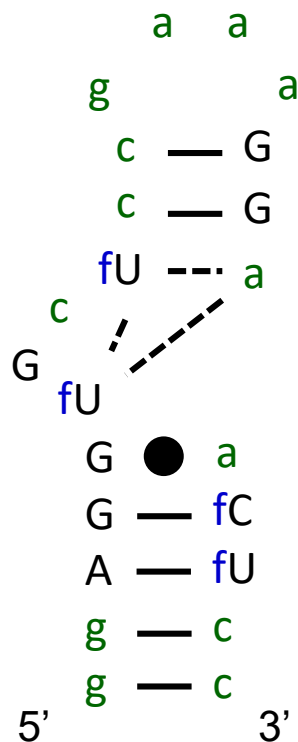
## In culture medium



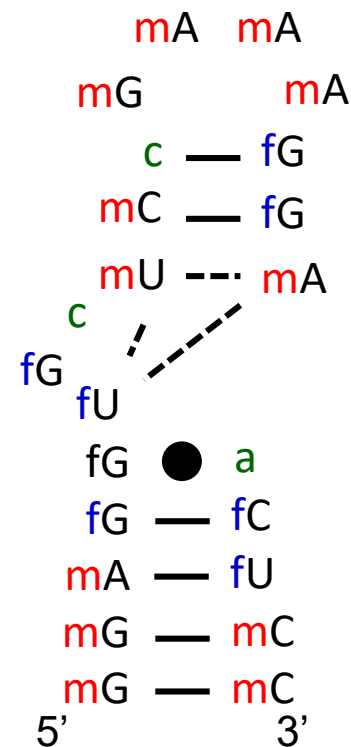
IgG-expressing cell culture supernatant at 37°C

# Secondary Structures of RBM-101 and its Prototype Aptamer

## Prototype

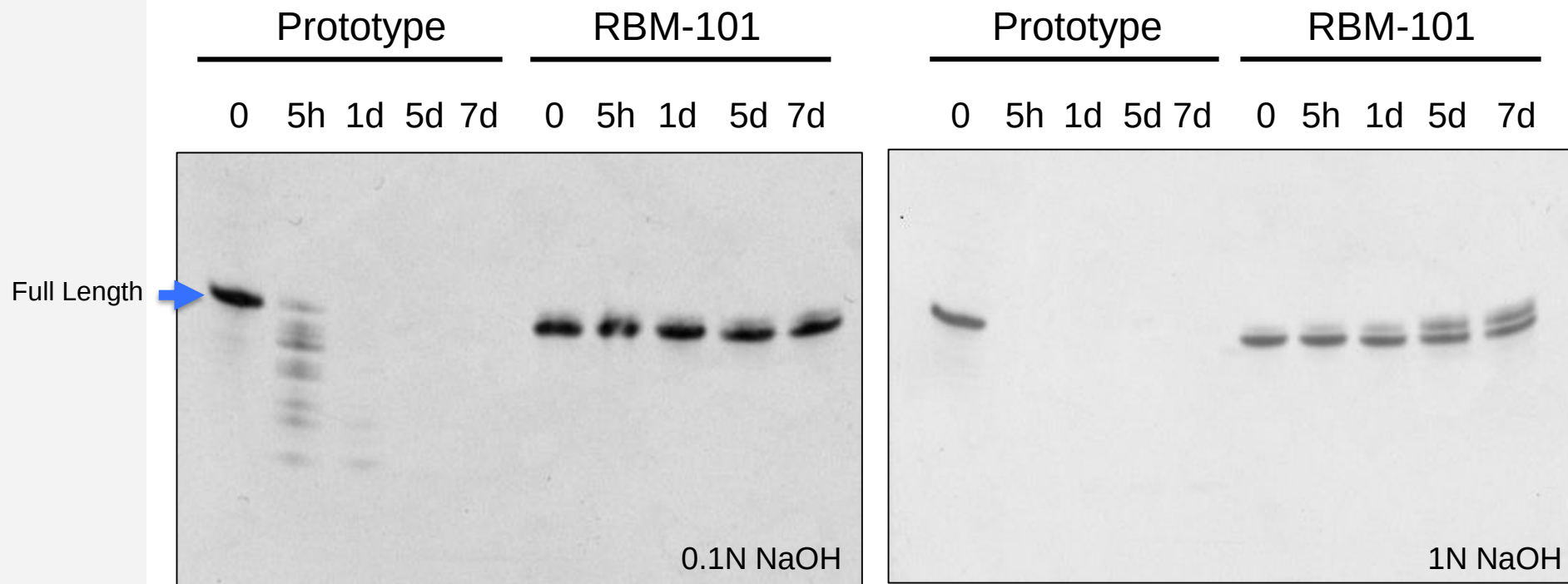


## RBM-101



*Biochimie. 2018, 145:113-124*

# RBM-101 is Highly Stable under Alkaline Condition

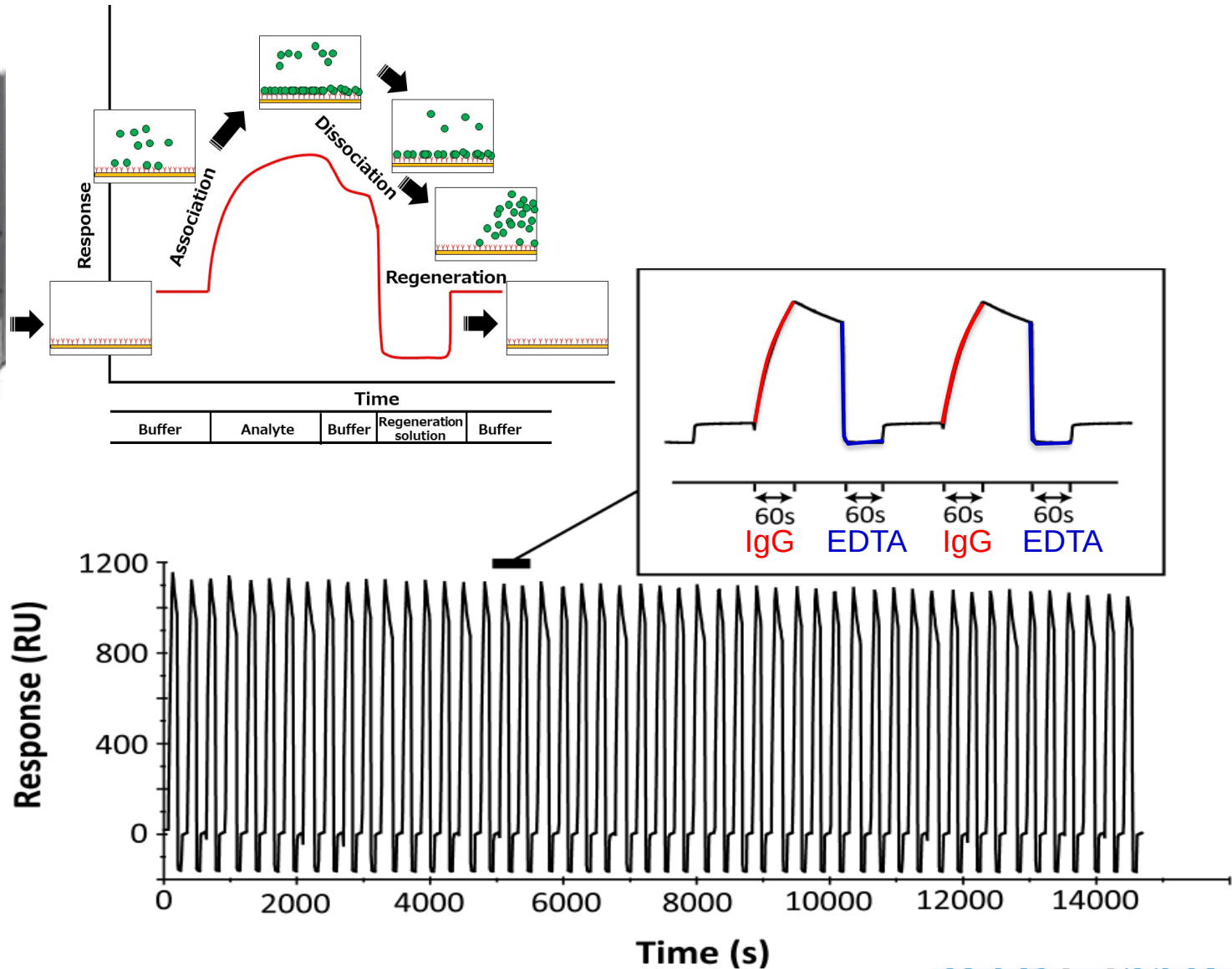


RBM-101 has no natural RNA residues and is highly stable under alkaline condition compared with the prototype aptamer that has some natural RNA residues.

# Repeated Regeneration of Aptamer



Biacore T200



# Affinity Profile of RBM-101

|         | Ka (1/Ms)          | Kd (1/s)              | K <sub>D</sub> |
|---------|--------------------|-----------------------|----------------|
| RBM-101 | $1.89 \times 10^5$ | $9.68 \times 10^{-3}$ | 51.2 nM        |

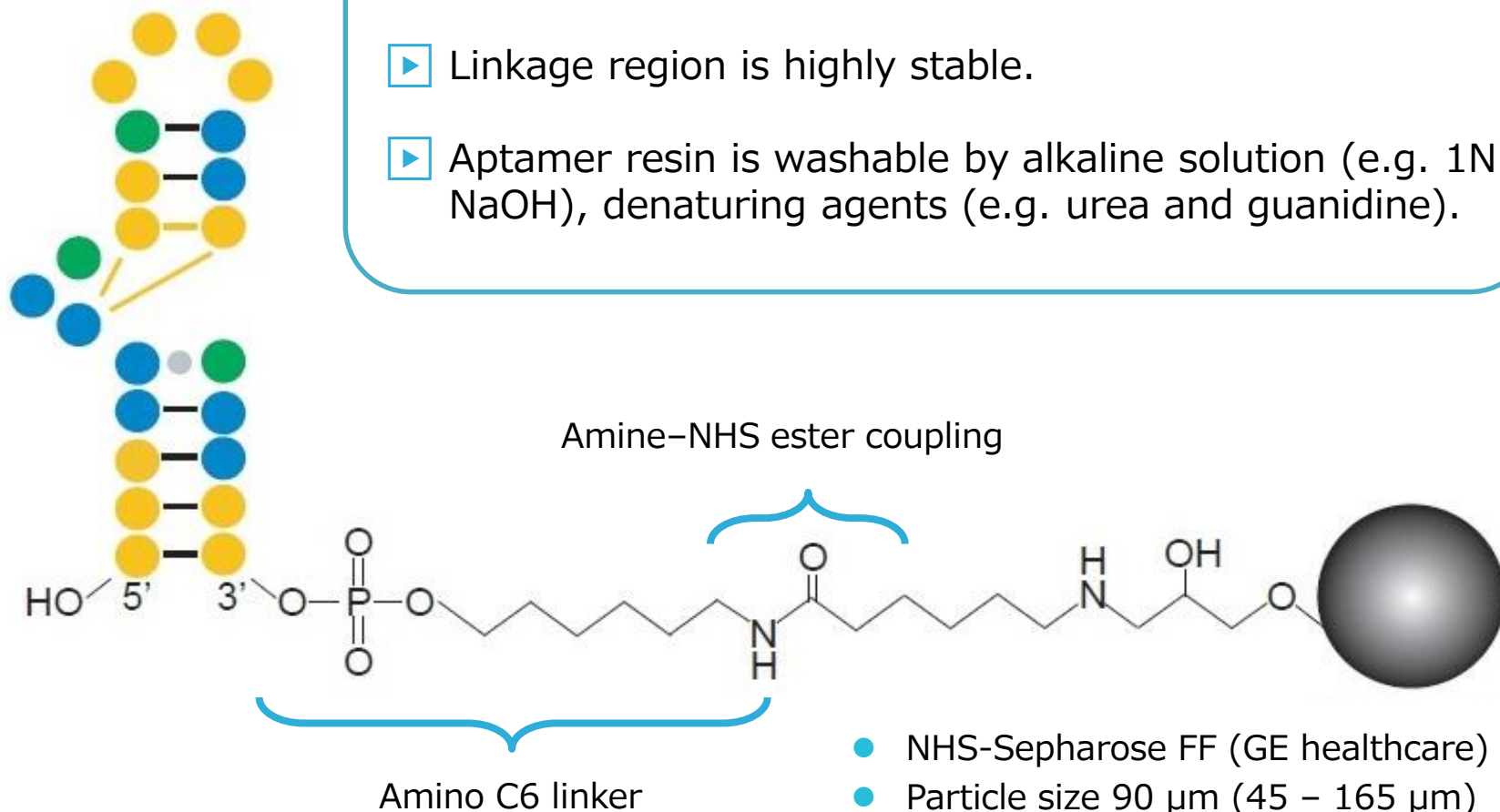
RBM-101 has enough affinity for purification of antibodies.

# Agenda

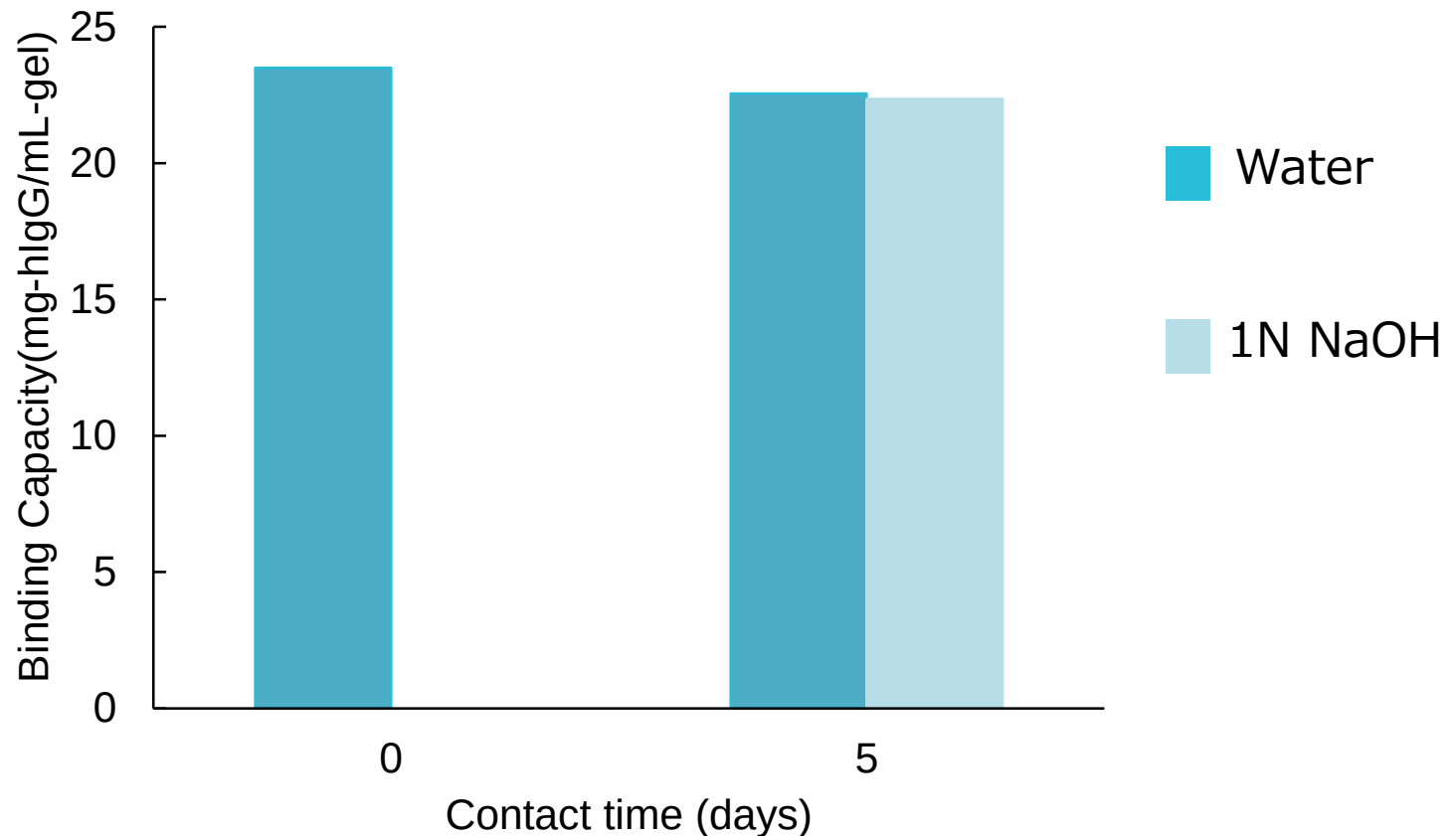
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# RBM-101 Immobilized Resin

- ▶ Aptamer can be modified on it's 5' and 3' terminus.
- ▶ Linkage region is highly stable.
- ▶ Aptamer resin is washable by alkaline solution (e.g. 1N NaOH), denaturing agents (e.g. urea and guanidine).



# RBM-101 Resin is Stable



After 5-days incubation with 1N NaOH at 25°C, RBM-101-immobilized resin maintains its binding activity.



# Specification of RBM-101 Spin Column

|                  |                         |
|------------------|-------------------------|
| Ligand           | Aptamer (Nucleic acid)  |
| Gel matrix       | Agarose                 |
| Binding capacity | > 4 mg human IgG/column |
| Support          | 100 µL                  |
| Column Volume    | 800 µL                  |
| Storage          | 4°C, Protect from light |

# Procedure of IgG Purification by RBM-101 Spin Column

## Remove preservation solution

Centrifuge for 10 sec.

## Equilibration

Add 600  $\mu\text{L}$  of 1 x binding / wash buffer, centrifuge for 10 sec.

## Binding

Place a cap on the bottom of a spin column, add pretreated sample, mix by inverting the spin column for 4 min, remove the cap and centrifuge 10 sec.

## Wash

600  $\mu\text{L}$  of 1 x binding /wash buffer, centrifuge for 10 sec. Repeat this step 3 times.

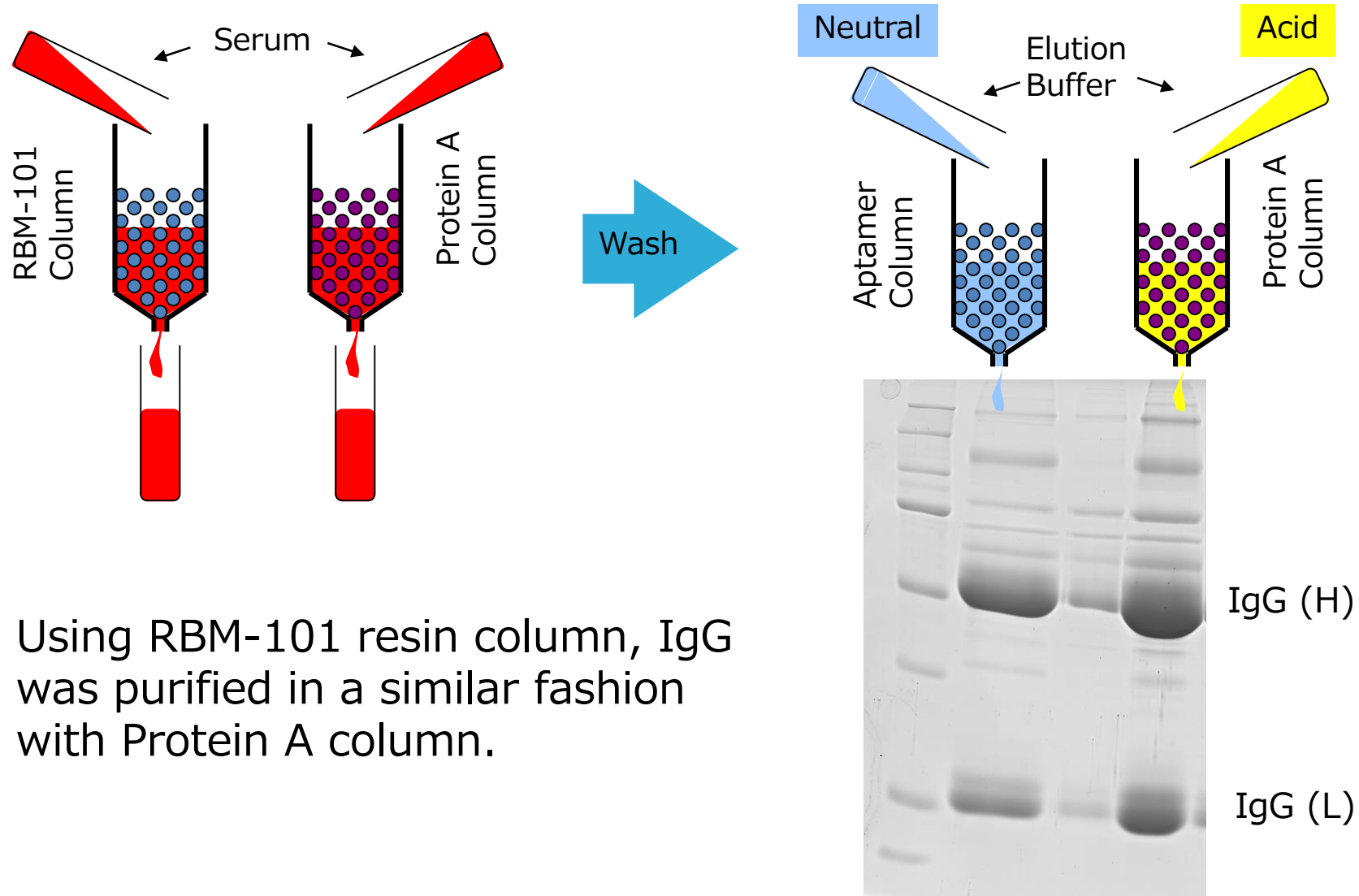
## Elution

Place a cap on the bottom of a spin column, add 600  $\mu\text{L}$  of **elution buffer containing EDTA**, mix by inverting the spin column for 4 min, remove the cap and centrifuge 10 min to collect eluting material. Repeat this step 3 times.

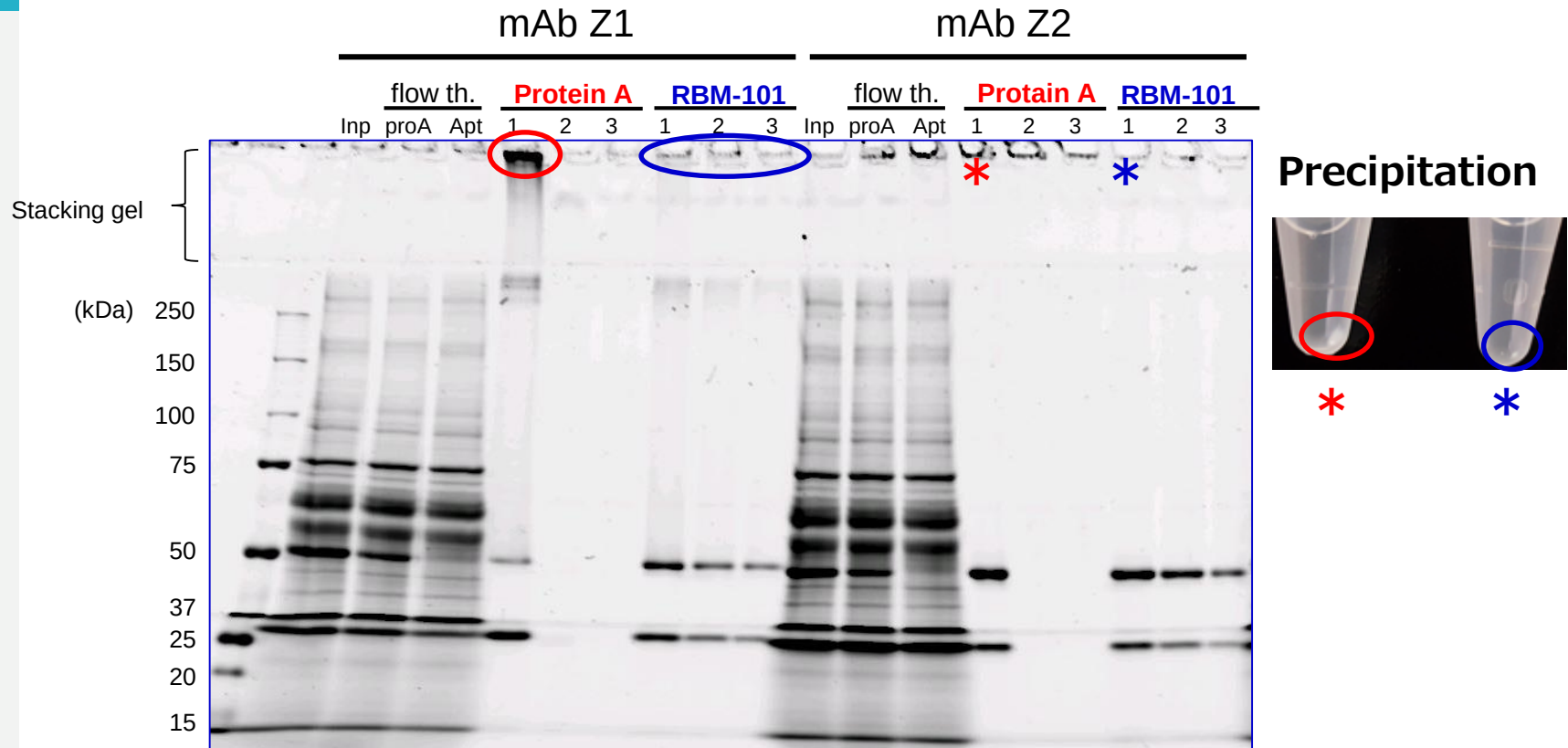
## Cleaning

Run 5 column volumes of 0.1 M NaOH.

# Case Study 1: IgG Purification from Serum



# Case Study 2: Antibody Purification from Cell Culture

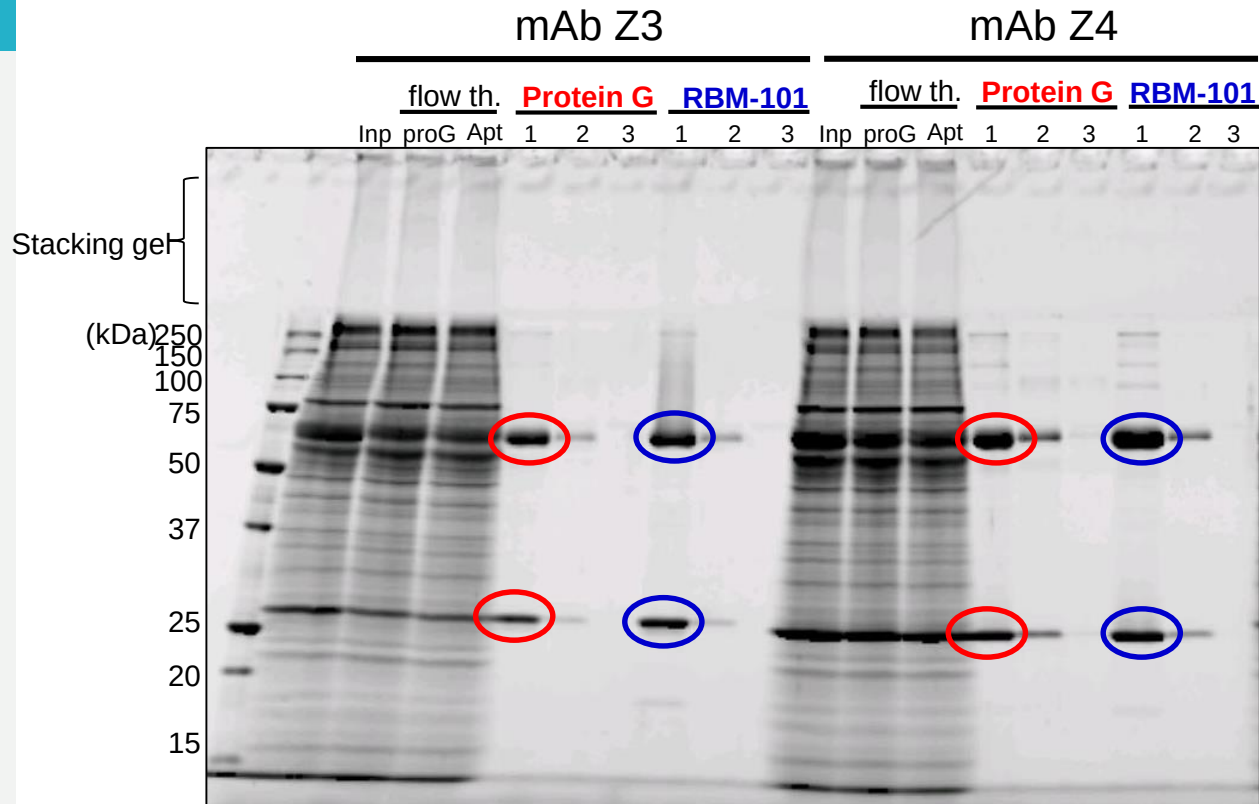


There are some cases where IgGs are aggregated in acidic elution from Protein A column.

Using Aptamer-immobilized column, such fragile IgG can be purified without generating aggregate during elution step under neutral condition.

# Case Study 3:

## Antibody Purification from Cell Culture



| Antibody | Resin     | Yield % | Yield % (After centrifugation) |                  |
|----------|-----------|---------|--------------------------------|------------------|
| mAb Z3   | RBM-101   | 54.7    | stable                         | 49.5             |
|          | Protein G | 53.9    | aggregation                    | 25.8 (decreased) |
| mAb Z4   | RBM-101   | 58.5    | stable                         | 49.9             |
|          | Protein G | 48.3    | aggregation                    | 16.4 (decreased) |

# Specification of RBM-101 LC Column for AKTA System



ÄKTAexplorer  
GE healthcare home page

|                          |                                 |
|--------------------------|---------------------------------|
| Gel matrix               | Agarose                         |
| Binding capacity         | 20 - 40 mg human IgG/1 mL-resin |
| Storage (long term)      | 20% EtOH                        |
| Maximum linear flow rate | 150 cm/h                        |
| Pressure                 | 0.1 MPa                         |

# Procedure of IgG Purification by RBM-101 LC Column

## Equilibration

Wash with five column volume of 1 x binding/wash buffer

## Binding

Inject pre-treated sample

## Wash

Wash with five column volumes of 1 x binding/wash buffer

## Elution

Elute with five to ten volumes of elution buffer at r.t.

## Cleaning

Run more than five column volumes of 0.1 M NaOH

## Storage

Run more than five column volumes of 20% EtOH

# Case Study: Purification with AKTA System

## Purification of human antibody X with AKTA system

| Column           | Treatment of eluate | Yield % of antibody X | Aggregation (Visual check) |
|------------------|---------------------|-----------------------|----------------------------|
| RBM-101          | Before dialysis     | 41.1                  | No                         |
|                  | After dialysis      | 37.2                  | No                         |
| Hitrap Protein G | Before dialysis     | 25.2                  | Yes                        |
|                  | After dialysis      | 20.6                  | Yes                        |

- RBM-101 resin is compatible with AKTA system.
- Antibody can be purified by AKTA system using RBM-101 column without generating aggregation.





# Summary:

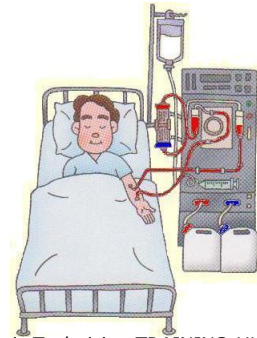
## Comparison of RBM-101 with Protein A

|                         | RBM-101           | Protein A    |
|-------------------------|-------------------|--------------|
| Elution pH              | Neutral           | Acidic       |
| Acid-sensitive antibody | Good              | No good      |
| Specificity             | High              | Low          |
| Purity                  | High              | High         |
| Capacity                | High              | High         |
| Stability               | Very Good         | Good         |
| Chemical synthesis      | Yes               | No           |
| Cost                    | Low (near future) | Leveling-off |

# Potential Therapeutic Applications of RBM-101 for Autoimmune Diseases

## Autoimmune diseases

- ▶ Caused by bankruptcy of immuno-tolerance.
- ▶ Rheumatoid arthritis, type 1 diabetes, basedow disease, ulcerative colitis, Crohn's disease, systemic lupus erythematosus, myasthenia gravis, immune thrombocytopenia, etc.
- ▶ 22 million patients in USA and 1 million patients in Japan.
- ▶ Patients are treated with steroids, immunosuppressive agents, thymus and/or spleen removed, plasma exchange therapy, etc.



Dialysis Technician TRAINING HUB  
home page



multiFiltratePRO  
Fresenius Medical Care  
home page



Medical device utilizing aptamer-immobilized membrane could be a good tool to treat these diseases to eliminate IgGs from patient's blood.

# Broad Potential Applications of RBM-101

- 1) Purification of human IgG<sub>1-4</sub> and Fc-fusion protein
  - for industrial manufacture of antibody therapeutics
  - for experimental use
- 2) Medical device such as a dialysis machine for the treatment of auto-immune diseases